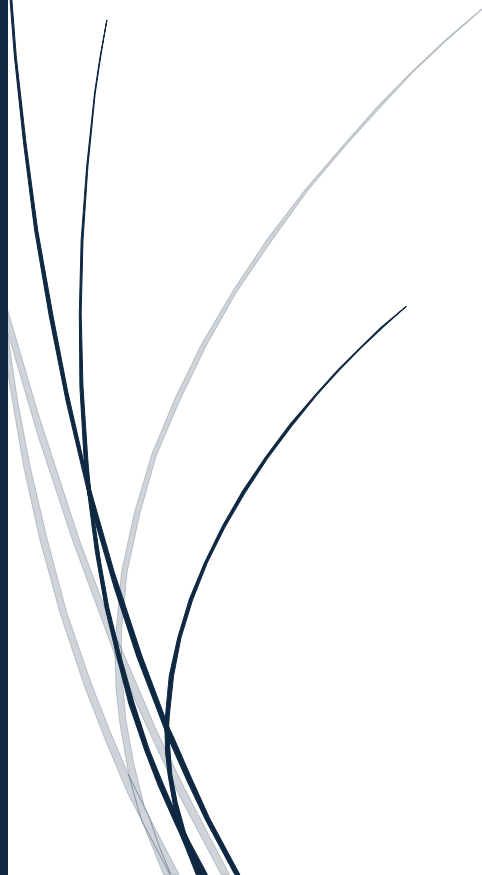


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Data Analysis using SQL

DataBase Assignment 3



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Task 1: Data Uploading & Cleaning

Data Uploading and Cleaning

- First, we created a table without specifying any primary or foreign key.
- Then we used the “BULK INSERT” method to upload data from “csv” files to our created tables.
- Next, we run the following query:

```
SELECT COUNT (*)  
FROM table t  
GROUP BY t.column  
HAVING COUNT (*)>1
```

If this query gives a result, it means our selected column is not qualified as a primary key. If it does not give any results, it means our selected column qualifies as a primary key.

- If single columns fail to uniquely identify a row, we switch to “GROUP BY” pairs of columns. Eventually we identify our primary keys.
- Then the previous table is dropped and again created by specifying the primary key.

Primary Key of Geolocation

No columns of the geolocation table qualified as a primary key. We made an extra column named “zip_code”, an auto-number integer, which we kept as our primary key. Due to this change, we had to add another heading in the csv as well because BULK INSERT inserts data based on the same number of columns and attributes in csv and table respectively. Also, the foreign keys in the other tables were not made at the time of creation. Later we used:

```
ALTER TABLE customers ADD zip_code int
```

Which adds a column to our customers table. Now we want to add zip_code values from the geolocation table based on the already present column customer_zip_code_prefix from the corresponding geolocation_zip_code_prefix which is why we used:

```
UPDATE c  
SET c.zip_code = g.zip_code  
FROM customers c
```

```
JOIN geolocation g ON c.customer_zip_code_prefix =  
g.geolocation_zip_code_prefix
```

Now we have a column of customers that can act as a foreign key from geolocation zip_code. To add the foreign key constraint:

```
ALTER TABLE customers  
ADD CONSTRAINT FK_customers_geolocation  
FOREIGN KEY (zip_code) REFERENCES geolocation (zip_code)
```

Similarly, we added the foreign key in the sellers table as well.

Changes made for Unique Identification

Based on our above approach, following were the primary keys that were identified.

Single Column Primary Keys:

- customers: customer_id
- orders: order_id
- products: product_id
- sellers: seller_id
- geolocation: zip_code

Two Column Primary Keys:

- order_reviews: review_id, order_id
- order_payment: order_id, payment_sequential
- order_items: order_id, order_item_id

Task 2: Data Retrieval (SQL Analysis)

Order Analysis

- a. Percentage of orders that got delayed beyond the estimated date.

Query Result Summary:

The percentage of orders that were delayed is 7.87099%.

b. What are the peak months of order delays?

Query Result Summary:

April is the peak month with the maximum number of orders delayed, which is 1565.

c. Which state experiences the highest order delays?

Query Result Summary:

SP state experienced the highest number of delays with a total of 2387 delayed orders.

d. See how many orders are still in “pending” status for each year.

Query Result Summary:

Orders that are still in “pending” state are show in in a table for each year. Not any specific pattern is observed. The highest number of pending orders are observed in the year 2017.

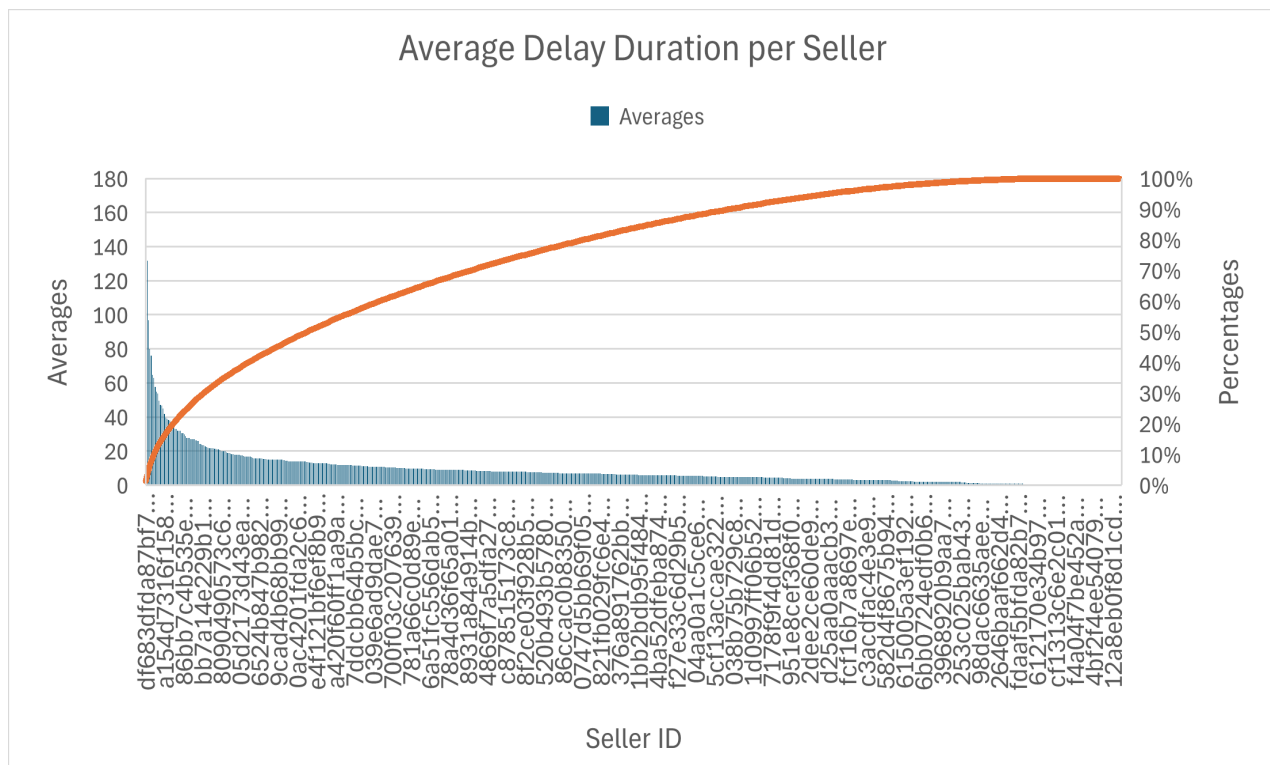
e. What is the average delay duration per seller?

Query Result Summary:

Highest average Delay duration is 167 days of seller id

“df683dfda87bf71ac3fc63063fba369d”. Only 5 sellers have an average delay of greater than 100 days.

Graphical Representation:



Key Findings from Graph:

The graph shows that 20% of the sellers have longer average delay duration, more than 40 days, including some having delays longer than 100 days. The rest of 80% of sellers have delays in less than one month.

- f. How do shipping costs impact order delays? Find the average shipping cost for delayed and on-time orders.

Query Result Summary:

According to the data observed, the delayed orders tend to have a higher shipping cost than on-time orders. This could indicate that expensive products are more likely to have delays in their deliveries than on-time products.

- g. Which product category experiences the most order delays?

Query Result Summary:

The product category cama_mesa_banho experienced the highest number of order delays of 920.

- h. How do the number of items per order affect the delays? Find the average number of items for delayed and on-time orders.

Query Result Summary:

The results show that both delayed, and on-time orders have same number of average items, which means that the number of items do not significantly affect the time of order delivery.

Customer Analysis

- a. What percentage of customers have made only one order?

Query Result Summary:

96.88% of customers have placed only one order.

- b. Find the top five cities with the most repeat customers (customers who have placed orders more than once).

Query Result Summary:

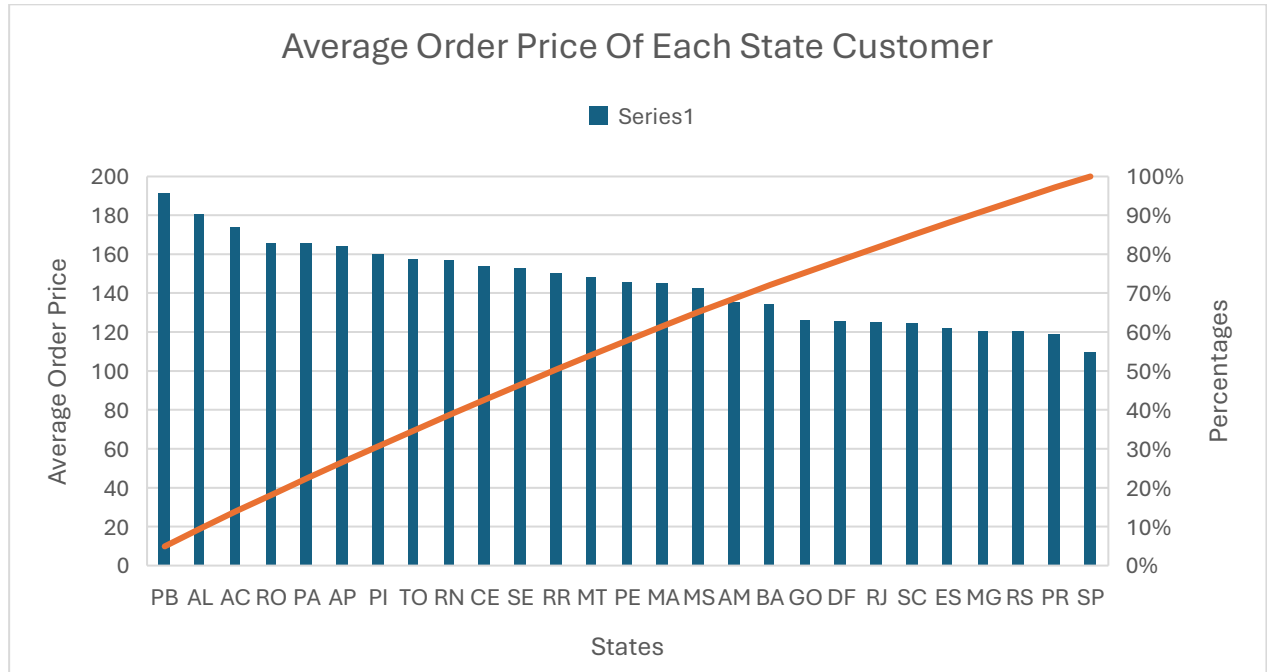
The top five cities are shown with the highest number of customers who have placed more than one order.

c. Calculate the average order price of customers for each state.

Query Result Summary:

The highest average price of order placed is 191.475\$ in the state PB and lowest is \$109.654 in the state SP.

Graphical Representation:



Key Findings from Graph:

Some states like PB, AL and AC have significantly higher average order prices, contributing effectively to the total revenue. Fewer states have low average order prices, which are low price states compared to others.

d. Find the top ten customers with the highest number of orders placed.

Query Result Summary:

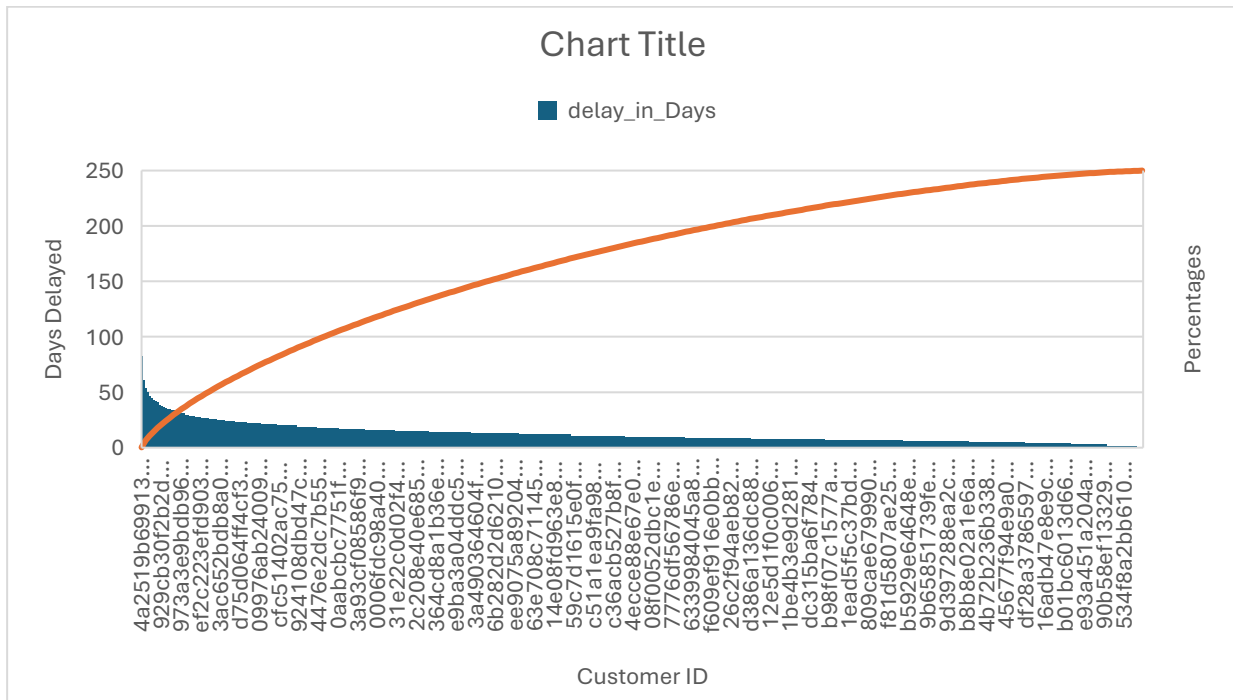
The top ten customers with the highest number of orders placed are displayed.

e. Which customers have the longest average delivery time?

Query Result Summary:

The highest delay time in days is 171 while only 4 customers have delay time in days greater than 100.

Graphical Representation:



Key Findings from Graph:

The graph shows that 20% of the customers have a longer average delivery time, more than one month, including very few having delays longer than 200 days. The rest of 80% of customers have delays in less than one month.

- f. How does customer order frequency change over time? Find the average number of orders placed per customer per year.

Query Result Summary:

The average number of orders placed per customer is same for each year, which shows that no change is observed in the customer order frequency over the three years (2016-2018).

- g. Which of the top 5 customers have spent the most money in year 2017?

Query Result Summary:

\$13664.080 is the highest amount spent by the customer in the year 2017 while the 5th highest is \$6081.540.

h. Which customers have the highest order cancellations?

Query Result Summary:

The customer with customer id “ef0103e9602d12594d19c2b666219bc1” has the highest number of order cancellations that is equal to 3 orders cancelled.

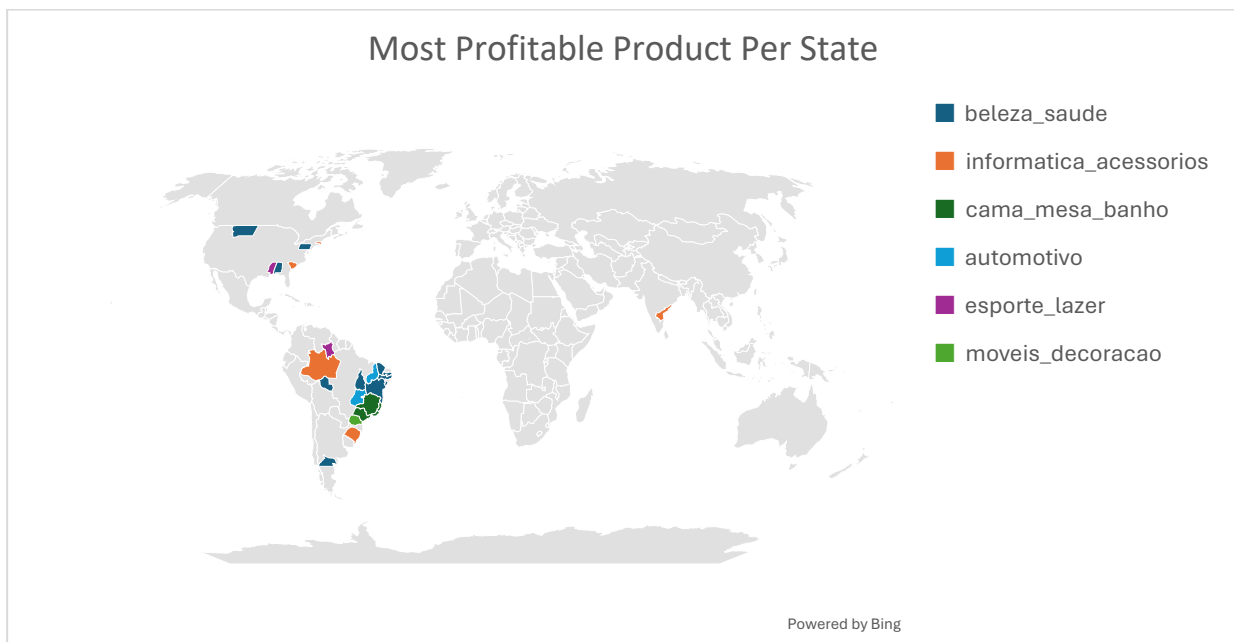
Product Analysis

a. What is the most profitable product category per state? Find the most profitable product category in each state based on total sales.

Query Result Summary:

Most profitable among states is automotivo, beleza_saude, informatica_acessorios etc.

Graphical Representation:



Key Findings from Graph:

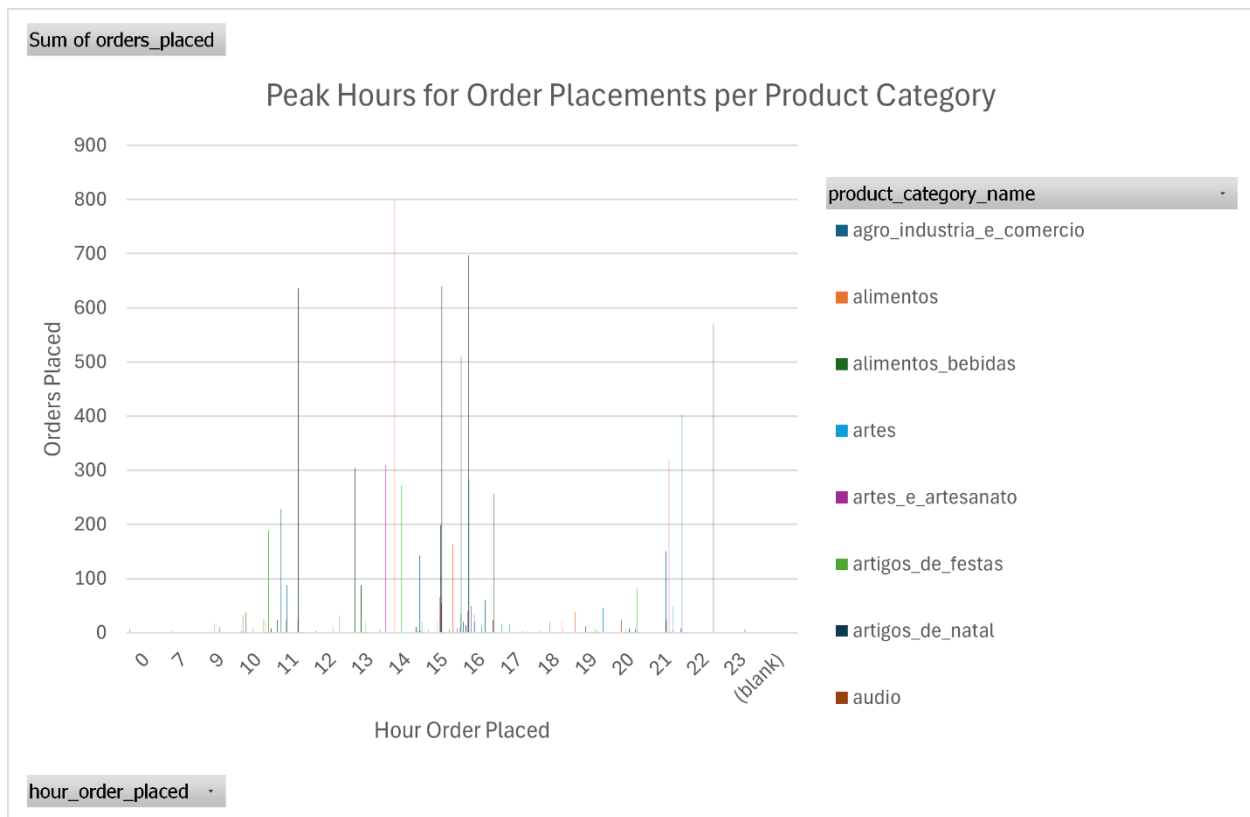
The graph shows that most profitable products are from the states in the eastern side of South America. Very few are from North America as well.

- b. What are the peak hours for order placements per product category?
Find which hours of the day have the highest number of order placements for each product category.

Query Result Summary:

For each product category, their peak hour has been showed in the 24-hour format (From 0th hour to 23rd hour), meaning the hour of the day in which maximum orders have been placed for that product category.

Graphical Representation:



Key Findings from Graph:

The graph shows that most of the peak hours for order replacement occurs at 11, 14, and mostly for the 16th hour of the day.

- c. Which product categories experience the most delays? Find the top 5 product categories with the highest number of delayed orders.

Query Result Summary:

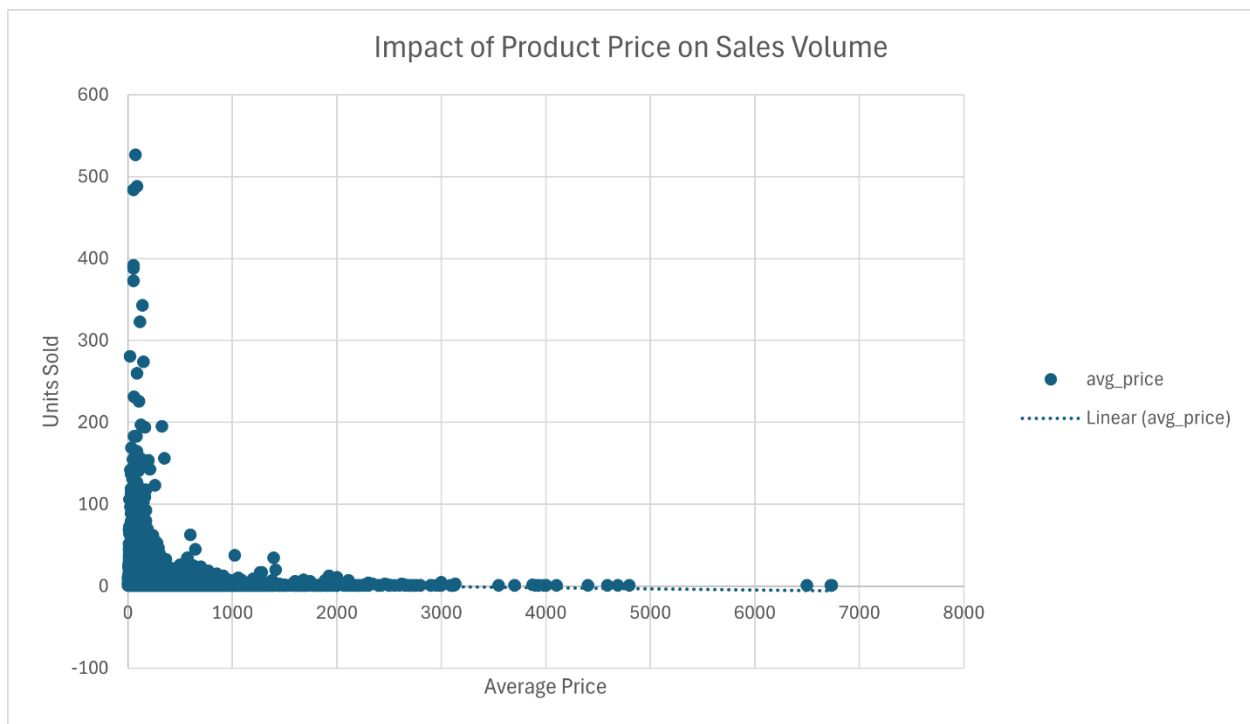
920 delayed orders for the category cama_mesa_banho while the 5th highest order delay is 594 items.

- d. What is the impact of product price on sales volume? Find whether higher-priced products sell less by comparing average price vs. total sales.

Query Result Summary:

According to the analysis of the data, most of the products with low prices have a high volume of sale, such as the product categories moveis_decoracao, cama_mesa_bahno, and many more. Meanwhile some products like “pcs”, “artes” and some others are high priced and have very smaller number of units sold. However, there may be some high-priced products with high sales due to customer demand.

Graphical Representation:



Key Findings from Graph:

The graph shows that lower price products are sold in a much more quantity than higher priced products. Higher priced products have a significantly low amount sold.

- e. Which products are most frequently bought together? Find the most frequently purchased product pairs.

Query Result Summary:

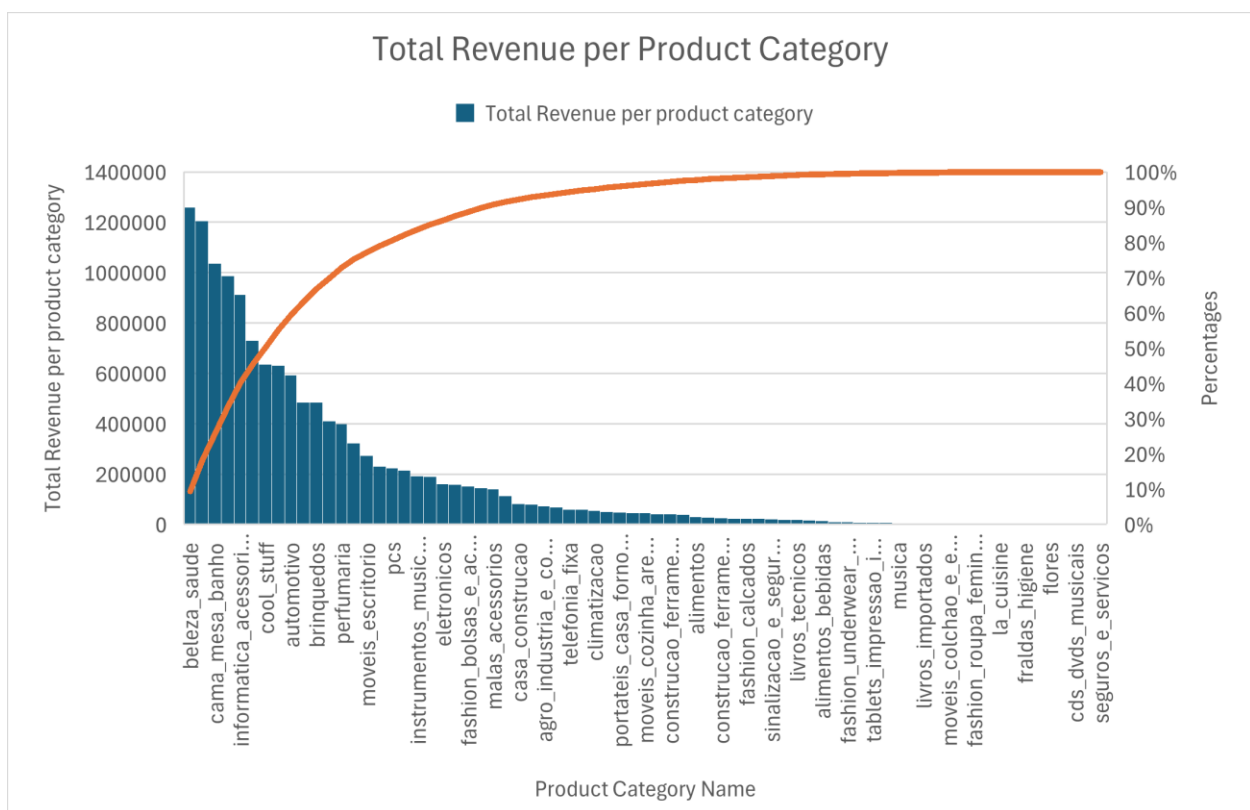
The pair of products (any two products) that have been bought in the same ordered for the greatest number of times are “cama_mesa_banho” and “moveis_decoracao” with a frequency of 128.

- f. Calculate the total revenue per product category by summing order item prices.

Query Result Summary:

The total revenue (total money earned by the company) for each product category is shown.

Graphical Representation:



Key Findings from Graph:

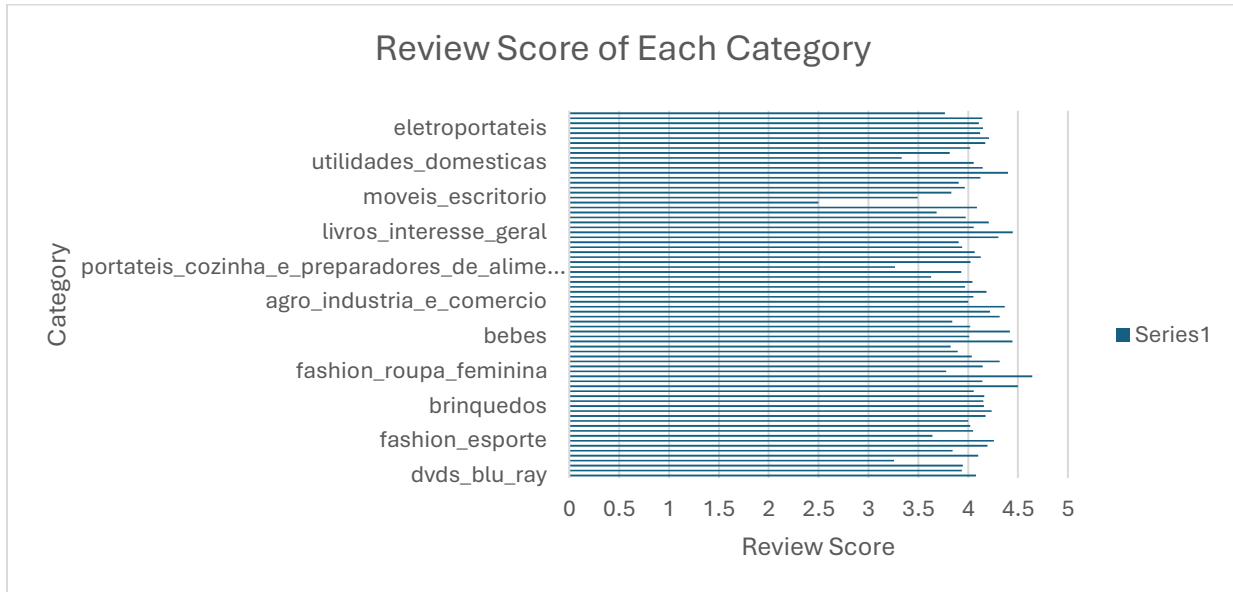
The graph shows some of the popular product categories (6-7 categories) having a high sales revenue higher than 600,000. The rest have sales revenue lower than that.

g. Compute the average review score for each product category.

Query Result Summary:

The average review score mostly is in the range of 3.0-4.0 for all the product categories with few outliers.

Graphical Representation:



Key Findings from Graph:

The graph shows that the product category with CDs, DVDs and musicals have the highest review score average. Others also have average review scores lying around 3-4.

h. Retrieve the top 5 products based on total sales revenue.

Query Result Summary:

The top five products are shown with the highest total sales revenue.

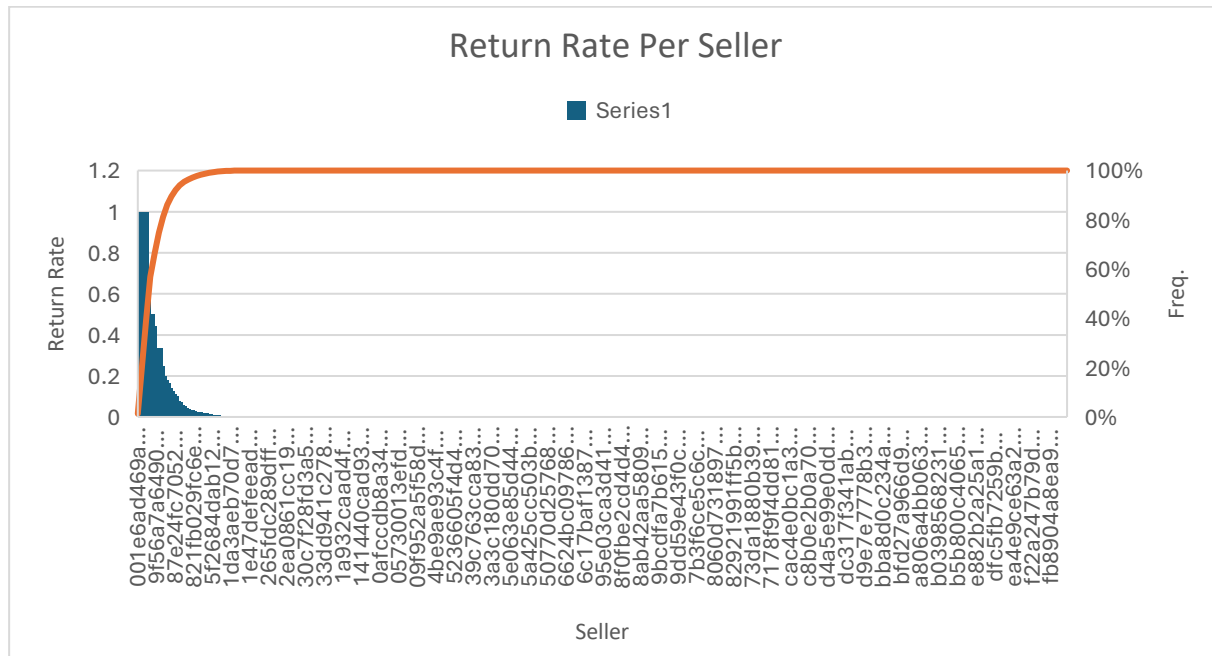
Seller and Shipment Analysis

a. What is the return rate per seller?

Query Result Summary:

The return rate is very high starting from 100%. More than 40 sellers have 100% return rate while others have less than 60% return rate.

Graphical Representation:



Key Findings from Graph:

The graph shows that the return rate is very high for a few sellers starting from 100%. More than 40 sellers have a 100% return rate while others have less than 60% return rate.

- b. Which sellers sell the most expensive products on average? Find sellers whose products have the highest average price.

Query Result Summary:

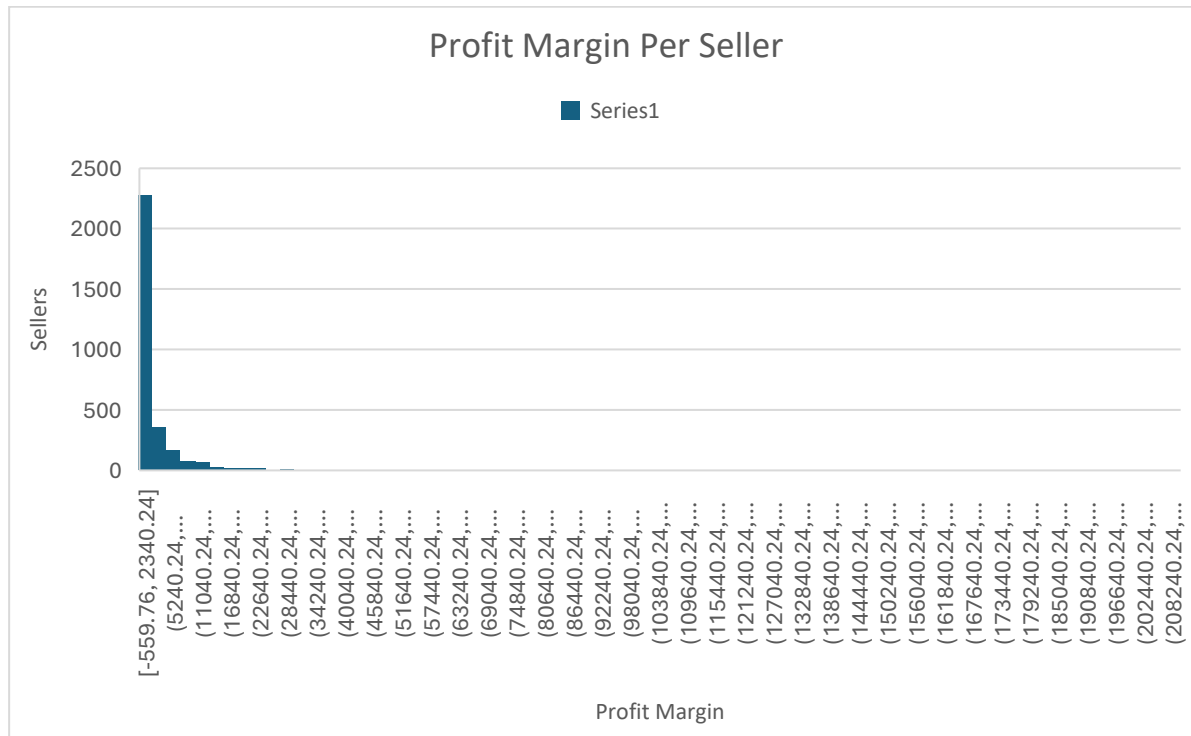
On average, the most expensive products are sold by the seller with seller_id “80ceebb4ee9b31afb6c6a916a574a1e2” with the average price of 6297.00.

- c. What is the profit margin per seller? Find the profit margin per seller, assuming:

Query Result Summary:

The top seller has a profit margin of 209695.420\$ with only 2 sellers above 2 lac profit margin.

Graphical Representation:



Key Findings from Graph:

The graph shows that very few sellers (2 sellers) have a profit margin higher than 200000. The rest have significantly lower profit margin compared to the highest.

- d. How do shipping costs impact order delays? Find the average shipping cost (freight_value) for delayed vs. non-delayed orders.

Query Result Summary:

According to the data observed, the delayed orders tend to have a higher shipping cost than on-time orders. This could indicate that expensive products are more likely to have delays in their deliveries than on-time products.

- e. What is the number of delayed shipments in 2017?

Query Result Summary:

Total of 4644 shipments are delayed in 2017.

- f. What is the correlation between shipping cost and delivery speed? Find the average shipping cost (freight_value) for delayed vs. on-time shipments.

Query Result Summary:

According to the data observed, the delayed orders tend to have a higher shipping cost than on-time orders. This could indicate that expensive products are more likely to have slow delivery speeds than on-time products.

- g. Sum the total freight cost for each seller.

Query Result Summary:

Highest freight cost is 51612.550\$ while all other sellers have less than 35K\$ cost.

Graphical Representation:



Key Findings from Graph:

The graph shows that the highest freight cost is \$51612.550 while all other sellers have less than \$35K cost.

Task 3: Presentation

Exporting and Visualizing Data

- The data was exported externally from the SQL Server system.
- An empty CSV file was created on MS Excel.
- The results of the queries were copied from the option “Copy with Headers”, which allowed us to copy the headers as well.
- The result was pasted in the CSV files.
- Then, the recommended charts were created on MS Excel and copied to the report for visualization.